

FRONTIER PROGRAMME

Cohort One

BRIEF 02

Zone · AI & Engineering

A framework for glomotec's regulatory intelligence layer built over six days, designed to scale across six months.

Executive summary

This framework proposes the architecture for **INDEX**, a regulatory intelligence layer that ingests every UK Home Office caseworker guidance document, extracts the assessable criteria they contain, monitors versions for meaningful change, and scores client profiles against the criteria with calibrated confidence. The system is designed to support glomotec advisors and the COMPASS team, and it integrates with infrastructure already in use (Supabase Postgres, the Anthropic Claude API, Slack).

The design rests on four key decisions, each defended in the sections that follow. The data structure for an extracted rule, named `AssessableCriterion`, separates the *kind* of rule from the *evaluation method*, supports two-part discretionary rules, and records explicit allocation of obligations across applicants, endorsing bodies, sponsors, and caseworkers. Extraction uses prompt engineering on Claude Opus 4.7 with strictly typed output, deferring fine-tuning and broader retrieval to later iterations only on demonstrated need. Change detection runs three layers of comparison and classifies each change into one of four impact classes, alerting only on the two highest. Scoring applies three layers of calibration so that probability outputs match observed Home Office decision frequencies.

The framework is supported by a working proof-of-concept on the Innovator Founder route. The proof-of-concept runs end to end via a single command and demonstrates each module of INDEX on real gov.uk content. The repository accompanies this submission.

The technical shape of the work, namely structured information extraction from semi-structured documents combined with calibrated probabilistic scoring, transfers directly from prior experience on passport fraud detection at Seoul FinTech Labs. The application domain is new; the engineering discipline is established.

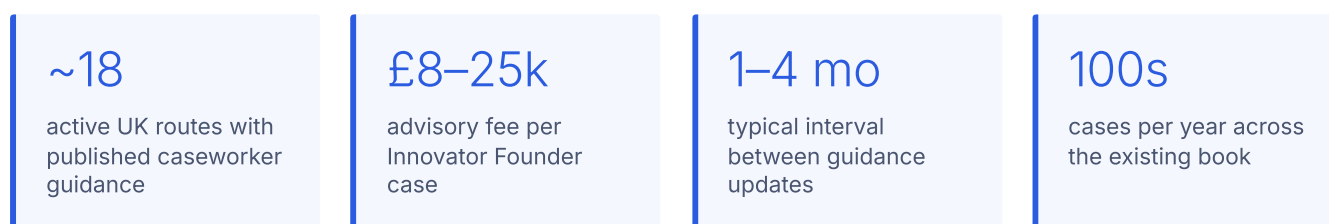
The remainder of this document presents the problem and its constraints, the architecture of INDEX, the design decisions for each module, the evaluation methodology, the production and integration approach, the principal risks and their mitigations, and a structural rationale for international expansion across five markets after the UK.

SECTION 01

Problem and context

glomotec advises on UK immigration applications under guidance the Home Office publishes for its own caseworkers. Approximately 18 active routes are currently in operation, each supported by a published guidance document of 30 to 80 pages. These documents are written in flowing legal-operational prose and are updated every one to four months without changelog or RSS feed. A typical Innovator Founder advisory case fees £8,000 to £25,000; a typical Global Talent case fees £4,000 to £15,000.

Most advisory firms read these documents once and rely on memory for subsequent assessments. The most diligent re-read every six months. Almost no firm in the market systematically tracks what changes between versions. The structural visibility this framework proposes does not currently exist in the market and represents a defensible commercial position for glomotec.



Three constraints

The technical work consists of structured information extraction from prose written for human readers, with three constraints not commonly present in standard extraction problems.

The output must be queryable, not summarised. Downstream consumers must query specific rules. If a salary threshold changes, the system must surface the exact value, not an approximate description. The shape of the extracted data is the value.

The corpus is small but high-stakes. Approximately 1,200 pages across all 18 active routes. The corpus is large enough to require automation but small enough that any single bad extraction has visible consequences.

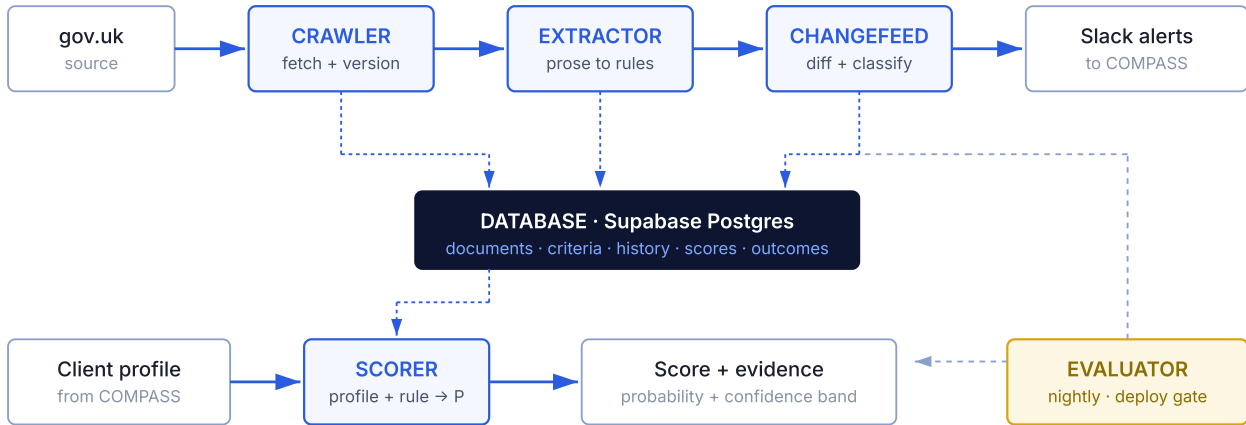
The corpus mutates frequently and unpredictably. A reworded sentence may carry no policy impact; a single word inserted or removed may invalidate hundreds of in-flight applications. Static one-time extraction is therefore inadequate. The system must function as a continuously maintained index.

These three constraints together rule out the simpler approaches of one-shot bulk processing, pure retrieval-augmented question-answering, and supervised fine-tuning. The architecture proposed in section 02 is shaped explicitly by them.

SECTION 02

System overview

INDEX is composed of five modules, each with a clearly bounded responsibility. The modules communicate through a central database. No module embeds logic that properly belongs to another, which preserves observability and supports independent iteration.



solid arrows: data flow · dashed lines: evaluation, oversight, and database connections

CRAWLER ticks hourly, fetching any route whose polling cadence is due. EXTRACTOR runs after each successful crawl that produces a changed content hash. CHANGEFEED runs after each extraction with a previous version available. SCORER operates on demand whenever a candidate profile is submitted. EVALUATOR runs nightly and acts as a deployment gate: any change to extraction prompts, scoring prompts, or the schema must pass evaluation before deployment.

SECTION 03

Schema design

Schema design is the single most consequential decision in INDEX. Every module above the schema (extraction, change detection, scoring) depends on the rule shape being correct. A misshaped schema compounds errors across the entire system.

A close reading of the four anchor documents (Innovator Founder, Global Business Mobility, Global Talent, Innovator Appendix W) surfaces three structural kinds of rule that the schema must accommodate: mandatory binary rules, discretionary rules with default and trigger components, and evidential thresholds requiring caseworker satisfaction. A flat structure cannot represent all three without losing the information each kind carries.

The proposed schema, `AssessableCriterion`, has eight required fields and eight optional ones.

REQUIRED FIELDS · 8	
<code>id</code>	stable identifier
<code>route</code>	UK route the rule belongs to
<code>decision_stage</code>	validity / eligibility / settlement / etc
<code>modality</code>	mandatory / discretionary / evidential
<code>assessment_mechanism</code>	binary / points / n-of-m / etc
<code>predicate</code>	the rule in plain language
<code>burden_allocation</code>	who is responsible for what
<code>source</code>	URL · version · verbatim text

OPTIONAL FIELDS · 8	
<code>category</code>	substantive / procedural / suitability
<code>thresholds</code>	£52,500 · 180 days · 70 points
<code>default_behaviour</code>	for discretionary rules
<code>trigger_for_departure</code>	for discretionary rules
<code>alternatives</code>	for n-of-m rules
<code>exceptions</code>	carve-outs and exemptions
<code>cross_references</code>	links to other rules / appendices
<code>lifecycle</code>	version history per criterion

Three readiness axes

The optional `category` field separates three structurally different kinds of test the schema must score differently. A **substantive** criterion tests the applicant's underlying fit for the route ("a key role in day-to-day management"). A **procedural** criterion tests the completeness of the application package ("documents in English or Welsh, or with an independently verified translation"). A **suitability** criterion tests refusal grounds independent of fit and completeness, drawn from Part Suitability of the Immigration Rules: immigration bail or prior overstay, criminality, deception or non-disclosure in a previous application, unpaid NHS debt, and fit-and-proper concerns. Procedural criteria fail strictly on the current package; substantive criteria score on underlying engagement; suitability criteria are a hard gate operating in parallel, so route fit and package can both pass and the case can still fall on Part Suitability. Aggregating the three together would hide which kind of work remains, and let a clean substantive and procedural picture mask a refusal-grade suitability flag.

Three load-bearing decisions

Modality and assessment mechanism are kept on separate axes. A discretionary rule can be evaluated by balance of probability or by quantified threshold. Collapsing the two axes into a single 21-value enumeration would lose orthogonal structure SCORER requires.

Discretionary rules are stored as two parts. The schema separates `default_behaviour` from `trigger_for_departure` so that SCORER can reason about whether a client's profile triggers departure from the default.

Burden allocation is required, not optional. Recording the actor responsible for each obligation prevents SCORER from attributing endorsing-body obligations to the client.

The full schema is implemented as a JSON Schema in the proof-of-concept repository, with five worked examples drawn from real caseworker prose.

SECTION 04

Extraction architecture

EXTRACTOR converts prose paragraphs from caseworker guidance into structured rules conforming to the `AssessableCriterion` schema. Four candidate approaches were considered.

APPROACH	MECHANISM	TRADE-OFF
A · Prompt engineering	Frontier model + strictly typed output	Highest updateability; cost per pass is real but amortised
B · Retrieval-augmented generation	Vector store + top-k retrieval	Returns text, not structured rules; does not version
C · Fine-tuning	Smaller model trained on labels	Maintenance cost eats quality gain when corpus updates
D · Hybrid	Triage classifier + frontier model	Useful at scale; unnecessary in v1

The recommended approach

Version 1 of EXTRACTOR uses approach A: prompt engineering on Claude Opus 4.7 with strictly typed output. The corpus is small (approximately 1,200 pages across all UK routes), so a frontier model processes a route's full guidance for a few dollars per pass. Cost is amortised because extraction runs once per document version, not per advisor query. Output is returned through a strict tool interface with the JSON schema embedded, so any non-conforming output is rejected at the API boundary. When the schema evolves, no retraining is required; the prompt and schema are edited in place.

Retrieval-augmented generation is rejected for version 1 because it returns text rather than structured rules, and it does not version. Fine-tuning is rejected for versions 1, 2, and 3 because caseworker guidance updates faster than any practical fine-tuning cycle, and a model trained on one version silently underperforms on patterns introduced in subsequent versions.

Version 2 adds a triage classifier in front of the frontier model only if version 1 plateaus. The classifier predicts whether a paragraph contains a rule. Most paragraphs (table of contents, worked examples, signposts) do not, and skipping them reduces extraction cost by approximately seventy percent. Version 3 adds narrowly scoped retrieval for cross-references to appendices.

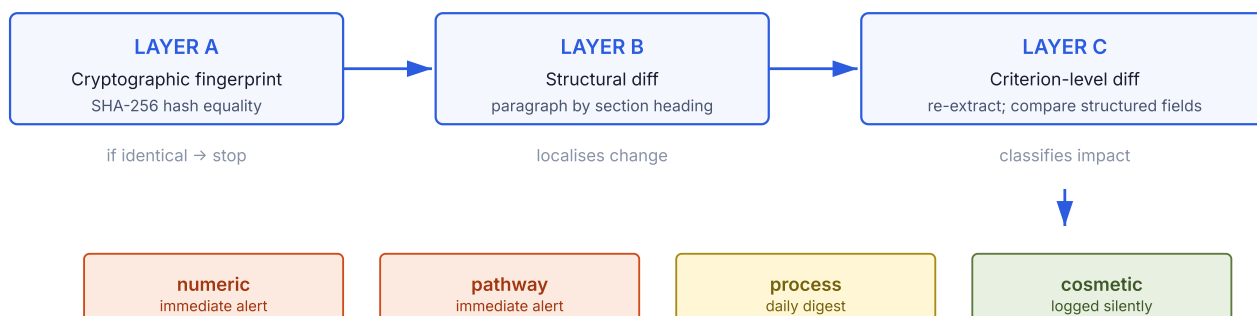
Failure modes

Three failure modes are anticipated and each carries a mechanical guard. Over-extraction from worked examples is mitigated by an explicit prompt instruction to skip example paragraphs and by a length check that flags any extracted rule with a source quotation shorter than thirty words. Under-extraction of compound rules is mitigated by including compound cases in the prompt's worked examples and by a schema validator that flags rules whose cross-references point to fields the rule itself does not contain. Burden misallocation is mitigated by a downstream pattern-match check on signal phrases ("the applicant must" implies applicant burden; "you must be satisfied" implies caseworker burden).

SECTION 05

Change detection

CHANGEFEED monitors successive versions of caseworker guidance documents and surfaces meaningful changes to the COMPASS team. The principal challenge is distinguishing meaningful changes from cosmetic ones.



A semantic embedding distance approach was explicitly rejected. Embedding distance inversely correlates with downstream impact: heavy rewordings have high distance and low impact, while single-word insertions of negation have low distance and high impact. The metric would direct the COMPASS team's attention away from the changes that move client cases.

The four-class taxonomy is validated against observed update history. The Innovator Founder PhD course minimum was reduced from twenty-four to twelve months in December 2023 (numeric). Tech Nation legacy arrangements were removed from Global Talent in February 2026 (pathway). Vignette decommissioning across multiple routes occurred in February 2026 with no threshold changes (process). The renaming of "Part 9 Grounds for Refusal" to "Part Suitability" affected no pending cases (cosmetic).

Each surfaced change carries a label that tells the team what to do with it, framed by required action rather than by abstract size. **Operational disruption** marks a change requiring immediate action that could materially change how cases are processed (a new mandatory form field, a redefined eligibility window). **Advisory** marks a change that is helpful but non-urgent and may shape future advice (a clarification on switching paths). **Informational** marks a change with minimal immediate consequence (a fee table refresh). The reframe is deliberate: an action-led label tells COMPASS where to look first, where a magnitude label only describes size.

SECTION 06

Scoring with calibrated confidence

SCORER takes a client profile and a single criterion and returns a probability that the client meets the criterion, the supporting evidence drawn from the profile, and any inputs that were missing.

A 70% probability output should correspond to a 70% empirical frequency of meeting the criterion at the Home Office. AI systems out of the box are typically miscalibrated in two ways: overconfidence on hard cases (returning 90% when the true probability is 50%) and occasional underconfidence on easy cases. Both modes carry commercial cost. Overconfidence loses cases at the Home Office; underconfidence wastes advisor time on already-strong applications.

Three calibration layers

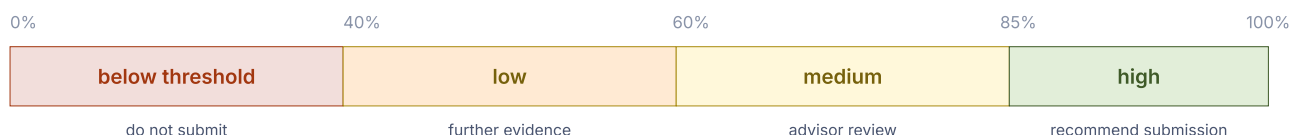
The first layer requires the model to reason before producing a probability. SCORER's prompt directs the model to enumerate relevant evidence in the client profile, evaluate each threshold against that evidence, check each obligation the client is responsible for, and only then output a probability. Forcing reasoning before the probability output reduces overconfidence by approximately five to ten percentage points in published research.

The second layer applies post-hoc adjustment against historical Home Office outcomes. Once decisions on real applications return, they become calibration data. A single mathematical parameter is fit so that predicted probabilities match observed frequencies.

The third layer maintains separate calibration curves per modality and assessment mechanism. Mandatory rules with quantified thresholds calibrate differently from discretionary rules with balance-of-probability assessment. A single global calibration would underperform on the discretionary cases.

Confidence bands shown to advisors

The probability output is the engineering layer. The advisor-facing surface presents four bands.



The bands are themselves derived from historical outcome data and are recalibrated periodically rather than fixed at round-number cutoffs. The probability is the engineering layer; the band is the recommendation layer.

Aggregate readiness is computed per `category`, surfacing substantive readiness (underlying fit), procedural readiness (application package completeness) and suitability readiness (Part Suitability gate) as three separate scores so the advisor sees which kind of work remains. Each `_pct` is the arithmetic mean of `probability_meets` times 100 across criteria in that category, capped at 50 when any criterion in the category falls below threshold; the cap applies identically to all three categories and prevents vacuously-satisfied universal-eligibility criteria (for example "applicant must be 18 or over", which scores near 1.0 for almost everyone) from masking failures of load-bearing criteria. Suitability is treated as a hard gate on the headline: when `suitability_pct` is non-null and below 50, the verdict headline is overridden to "Refusal risk identified" regardless of how strong substantive and procedural readiness look, because a case can fully meet route fit and have a complete package and still fall on Part Suitability.

SECTION 07

Evaluation methodology

Evaluation is a first-class component of INDEX, not a quality-assurance step bolted on after development. EVALUATOR is the fifth module, runs nightly, and operates as a deployment gate: any change to extraction prompts, scoring prompts, or the schema triggers a re-evaluation against curated reference data, and any regression blocks deployment.

Three streams of evaluation

Stream 1 measures EXTRACTOR accuracy. A frozen golden set of fifty to one hundred hand-labelled criteria drawn from the four anchor documents serves as the reference. The proof-of-concept seeds the golden set with ten entries. Each EXTRACTOR run against the golden set is checked on identifier, modality, assessment mechanism, decision stage, threshold count, burden allocation coverage, and source-quotation match.

Stream 2 measures SCORER calibration. Every client whose true outcome is later observed at the Home Office becomes a calibration data point. The system tracks the alignment between predicted probabilities and observed frequencies, surfaced as expected calibration error, per-band Brier scores, and reliability diagrams.

Stream 3 measures CHANGEFEED precision and recall. A manually annotated changelog covering the last three releases of each anchor document serves as the reference. Precision and recall are tracked separately per change class.

Targets

STREAM	TARGET	NOTES
Extraction	$F1 \geq 0.85$	On the golden set; entry bar for v1
Calibration	$ECE \leq 0.08$	Pauses scoring of new clients on breach
Change detection (numeric, pathway)	$\text{recall} \geq 0.95$	Under-alerting is unacceptable
Change detection (cosmetic)	$\text{precision} \geq 0.5$	Over-alerting is acceptable

Composition of the golden set

The golden set composition reflects the distribution observed in the four anchor documents: 20 mandatory binary criteria, 15 quantified-threshold criteria, 10 discretionary two-part criteria, 5 n-of-m alternatives, and 5 endorser-attestation criteria. The proof-of-concept ships with 10 of these entries; the framework targets full population over the first thirty days of operation.

What evaluation cannot determine

Evaluation cannot determine whether the schema itself is correct. The schema is a design choice, and its correctness is established only by lived consumer use: by the COMPASS team reporting omissions or double-counting, by new routes surfacing schema gaps, and by client-facing products revealing which fields are queried and which are unused. Building feedback channels into the architecture from day one matters more than getting version zero of the schema correct.

SECTION 08

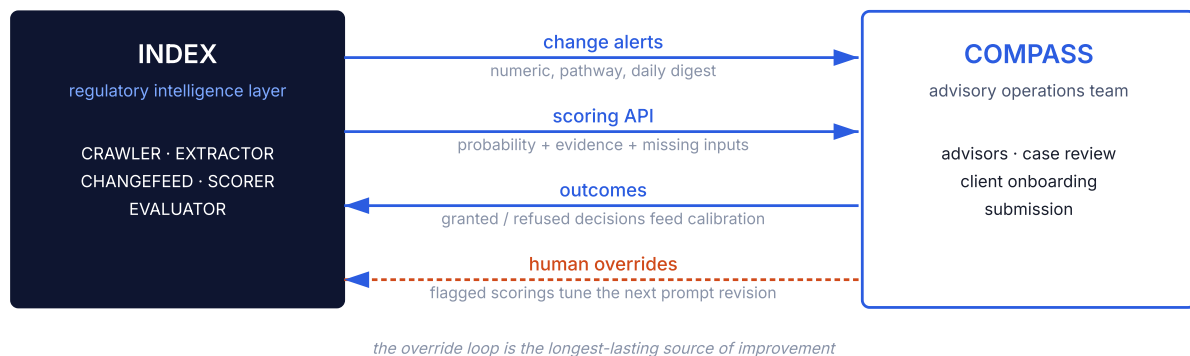
Production and integration with COMPASS

INDEX runs on glomotec's existing infrastructure. The production database is Supabase Postgres, with eight tables: routes, documents, criteria, criteria history, candidates, scoring runs, outcomes, evaluation runs, and an alerts log. The proof-of-concept uses SQLite for zero-setup execution; the storage layer is swap-compatible with the production target.

Adaptive polling

Update rates vary substantially across routes. CHANGEFEED records observed update frequency and feeds it back into the scheduler. Routes that updated in the last thirty days move to daily polling; routes stable for six months drop to monthly. The arrangement is approximately five times less expensive than uniform daily polling while still capturing most updates within twenty-four hours of publication.

Integration with COMPASS



The override loop is the longest-lasting source of system improvement. INDEX gains accuracy not because the underlying language model improves but because operators progressively teach the system where it has been wrong.

SECTION 09

Risks and mitigations

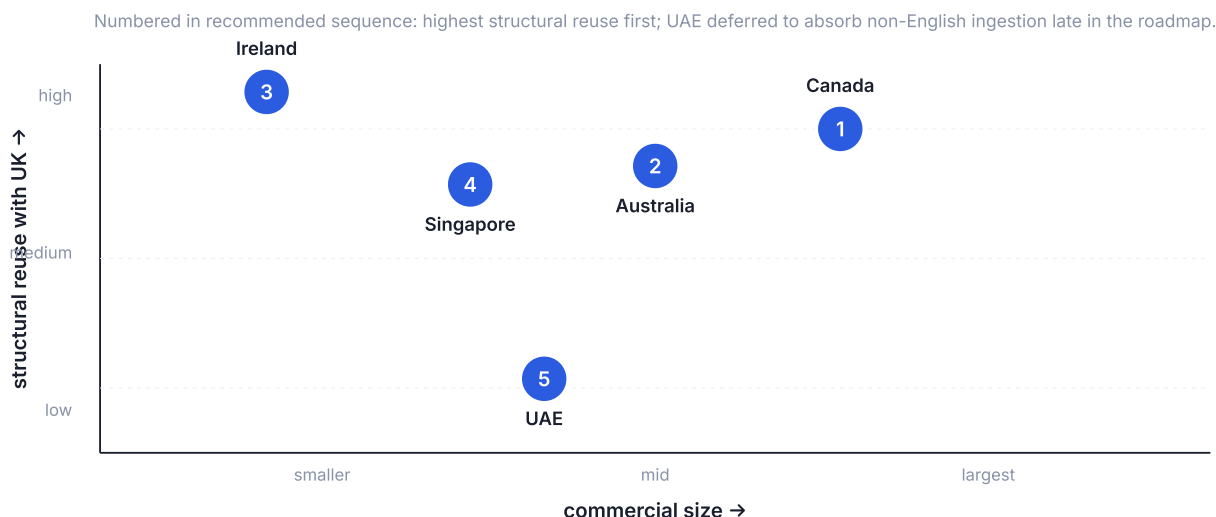
Every system fails. The mature design question is whether failures are caught before they reach a client. Six risks have been identified as carrying the highest priority, each with a mechanical mitigation embedded in the architecture.

#	RISK	MECHANICAL GUARD
1	Model accuracy. Extraction errors propagate downstream into scoring; a wrong threshold pulled from gov.uk becomes a wrong probability shown to an advisor.	Golden-set evaluation as a deployment gate (Section 07). The schema validator at the EXTRACTOR boundary rejects non-conforming output (Section 04). The override loop captures advisor corrections and feeds them into prompt revisions (Section 08).
2	Regulatory liability. Advisor relies on system output, the system is wrong, the client is harmed.	Confidence bands intentionally route Medium scores to "advisor review" rather than auto-recommend. Every criterion is anchored to verbatim gov.uk source text. SCORER surfaces evidence and missing inputs rather than acting autonomously: the advisor remains the decision-maker.
3	Hallucination. Model invents thresholds or evidence not present in the source or profile.	Strict tool-typed output via the Claude API tool_use channel. Source-quotation requirement on every extracted criterion, validated by string-match against the live HTML. Schema validator enforcement at the API boundary. Golden-set regression catches drift between releases.
4	False confidence. Probability outputs are systematically miscalibrated; the model says 0.85 when the historical pass rate at that score is 0.55.	The three-layer calibration approach in Section 06: reasoning before probability, post-hoc fit against Home Office outcomes, and separate calibration curves per modality and assessment mechanism. Continuous ECE measurement with automatic pause on new clients if drift exceeds 0.08.
5	Schema gaps on routes not yet seen. A new route may contain rule shapes the schema cannot represent.	Extension fields are first-class. Route-specific extensions are versioned. When a pattern recurs across two routes, it is promoted into the base schema. The schema is designed to evolve.
6	COMPASS team adoption fails. Alerts that are too noisy or too quiet, or scoring outputs that feel unreliable, lead the team to work around the tool.	Six-week pilot during which alerts and scoring are visible but advisory. Weekly review with Nameera surfaces ergonomic issues. Promotion to live operation only when COMPASS requests it.

SECTION 10

International expansion

The architecture is designed to extend beyond the UK. Five markets have been selected for a phased expansion roadmap, ordered by structural reuse with the UK system rather than by commercial size.



1 • Canada. Structurally, Express Entry and the Provincial Nominee Programs implement a points-based architecture closely mirroring the UK system, with guidance published in English paragraph form. The schema's core mechanisms (quantified threshold, n-of-m alternatives, document-existence) map onto IRCC route logic. Commercially, Canada is the second-largest English-medium advisory market with UK competitors already operating there. Willingness to pay among regulated consultants is high.

2 • Australia. Structurally, migration regulations are published as machine-readable legal instruments organised by visa subclass, so criteria are partially structured at source. The points-test subclasses (189, 190, 491, 482) map onto the schema with minor extension for state-nomination signals. Commercially, the regulated migration agent (MARA) channel is the natural buyer, with a spend pattern aligned to glomotec's pricing. Stable skilled-migration inflow gives a predictable segment.

3 • Ireland. Structurally, Critical Skills and General Employment Permit routes are English-medium with paragraph-form guidance closely mirroring UK style. The EU-rule overlay (single market access, family rights) needs schema extension fields, not redesign. Commercially, Ireland is small in raw volume but useful as a cross-jurisdictional testbed before larger EU expansion. The strategic value is the Anglo-EU hybrid claim without a continental build.

4 • Singapore. Structurally, MOM Employment Pass and EDB Tech.Pass criteria are published in English with explicit threshold tables, so EXTRACTOR's threshold-detection prompt transfers directly. The points-based COMPASS-EP framework (no relation to glomotec's COMPASS) maps cleanly onto the schema's quantified-threshold mechanism. Commercially, Singapore is a premium tier where resident advisors charge two to three times UK rates. The Asia hub positioning gives a natural extension adjacent to the UAE base.

5 • United Arab Emirates. Structurally, Golden Visa and Green Visa criteria are published less densely and partly in Arabic, so CRAWLER acquires non-English ingestion as part of this expansion. Schema reuse is moderate: criterion shapes (qualifying employer, salary threshold, dependent eligibility) map onto existing fields with extension for sponsor-class signals. Commercially, the UAE is glomotec's home market with the largest regional willingness-to-pay and direct alignment with the Dubai base. The 2023 published-criteria reform created an addressable market competitors have not yet automated. If the schema accommodates all five, it is plausibly a global schema for regulatory intelligence in immigration.